

AaharSetu – Smart Food Redistribution System

“Bridging Food to Lives”

1. Introduction

Food wastage is a major issue in urban areas where restaurants and events generate surplus food daily. At the same time, many people struggle to access basic meals.

AaharSetu is a technology-driven platform designed to bridge this gap by connecting restaurants, NGOs, and volunteers. The system ensures efficient, timely, and safe redistribution of surplus food.

2. Problem Statement

- Large amounts of food are wasted daily
- Lack of centralized coordination system
- Manual communication leads to delays
- No proper tracking of donations
- Food often spoils before reaching people

3. Objectives

- Reduce food wastage
- Improve coordination between stakeholders
- Ensure timely delivery of food
- Provide transparency and tracking
- Maximize social impact

4. Proposed Solution

AaharSetu is a centralized platform that:

- Connects restaurants, NGOs, and volunteers
- Enables real-time food donation and distribution
- Tracks the entire lifecycle of food delivery
- Ensures food safety through verification

5. System Overview

User Roles:

- 🍽️ Restaurant (Food Donor)
- 🤝 NGO (Receiver)
- 🚚 Volunteer (Delivery Agent)
- 🧑 Admin (System Manager)

Each user interacts through a dedicated dashboard.

6. System Workflow

1. Restaurant uploads surplus food
2. System notifies nearby NGOs
3. NGO accepts request
4. Volunteer is assigned
5. Food is picked up and delivered

6. Status is updated

7. Feedback is collected

Flow:

Restaurant → NGO → Volunteer → Delivery → Feedback

7. System Architecture

Architecture Type:

Client-Server Architecture

Components:

- Frontend: React.js dashboards
- Backend: Node.js APIs
- Database: Supabase
- APIs: Google Maps, Real-time services

8. Module Description

8.1 Restaurant Module

- Add food donation
- Track status
- View history

8.2 NGO Module

- View nearby food
- Accept/reject requests
- Track deliveries

8.3 Volunteer Module

- View tasks
- Accept delivery
- Update status

8.4 Admin Module

- Manage users
- Monitor system
- View analytics

9. Food Freshness System

AaharSetu uses a hybrid model:

- Time-based freshness tracking
- Food type rules
- NGO and volunteer verification
- Freshness score system

10. Technology Stack

- Frontend: React.js
- Database: Supabase
- APIs: Google Maps
- Real-time updates supported

11. MVP Features

- User authentication
- Food donation entry
- NGO acceptance
- Volunteer assignment
- Delivery tracking
- Basic dashboards

12. Advanced Features

- AI-based prediction
- Smart matching system
- Route optimization
- Impact analytics
- Emergency request system

13. Security & Trust System

- OTP-based authentication
- KYC verification
- Trust score system
- GPS tracking
- Proof of delivery

14. Limitations

- Dependency on internet
- Food safety challenges
- Volunteer availability
- Manual verification in MVP

15. Future Scope

- AI demand prediction
- IoT-based monitoring
- Blockchain transparency
- Mobile application
- Government integration

16. Impact

- Reduces food waste
- Helps needy communities
- Improves coordination
- Creates measurable social impact

17. Conclusion

AaharSetu is a scalable and impactful solution that transforms surplus food into a valuable resource. By leveraging technology, it ensures efficient redistribution and contributes to solving hunger and food waste challenges.